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# Deep Learning Prediction and Interpretation of Riverine Nitrate Export Across the Mississippi River Basin

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**Key Points:**

- Deep learning with high-frequency sensor data yields strong temporal forecasting predictions
- Interpretable machine learning identifies landscape drivers and thresholds controlling spatial variations in nitrate
- Export of surplus nitrate in the MRB is low overall but highly variable, driven by climate variability and legacy nitrogen

**Supporting Information:**

Supporting Information may be found in the online version of this article.

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**Abstract** Excess riverine nitrate causes downstream eutrophication, notably in the Gulf of Mexico where hypoxia is linked to nutrient-rich discharge from the Mississippi River Basin (MRB). We developed a long short-term memory (LSTM) model using high-frequency sensor data from across the conterminous US to predict daily nitrate concentrations, achieving strong temporal validation performance (median KGE = 0.60). Spatial validation—or prediction in unmonitored basins—yielded lower performance for nitrate concentration (median KGE = 0.18). Nonetheless, spatial validation was crucial in quantifying the impact of current data gaps and guiding the model's targeted application to the MRB where spatial validation performance was stronger (median KGE = 0.34). Modeling results for the MRB from 1980 to 2022 showed relatively low riverine nitrate export (19 ± 4% of surplus), indicating large-scale retention of surplus nitrate within the MRB. Interannual nitrate yields varied significantly, especially in Midwestern states like Iowa, where wet-year export fractions (42 ± 24%) far exceeded dry year export (6 ± 6%), suggesting increased hydrologic connectivity and remobilization of legacy nitrogen. Further evidence of legacy nitrate remobilization was noted in a subset of Midwestern basins where, on occasion, annual surplus export fractions exceeded 100%. Interpretable Shapley values identified key spatial drivers influencing mean nitrate concentrations—tile drainage, roadway density, wetland cover—and quantitative, non-linear thresholds in their influence, offering management targets. This study leverages machine learning and aquatic sensing to provide improved spatiotemporal predictions and insights into nitrate drivers, thresholds, and legacy impacts, offering valuable information for targeted nutrient management strategies in the MRB.

## 1. Introduction

Nitrate pollution leads to detrimental ecological effects, such as surface water eutrophication and groundwater contamination, which threaten both aquatic and human health (Galloway et al., 2008; Wurtsbaugh et al., 2019). In the United States, agricultural intensification and hydrologic alterations have been identified as leading causes of this water quality impairment, particularly in agriculturally intensive regions (Dubrovsky et al., 2010). This is most prominent in the Mississippi River Basin (MRB), where nitrate from the Corn Belt, a recognized pollution hotspot due to fertilizer over-application, contributes to a large nitrogen surplus (Stackpoole et al., 2021; Tian et al., 2020). This surplus, the difference between total nitrogen inputs and outputs like crop uptake (Byrnes et al., 2020; Sabo et al., 2019), is available for riverine export. Ultimately, the export of surplus nitrate from the MRB is a major contributor to the annual hypoxic "dead zone" in the Gulf of Mexico (Rabalais & Turner, 2019; Turner, 2024), which incurs an estimated yearly cost of $2.4 billion to fisheries and marine habitats (Boehm, 2020). While nitrate pollution has been well-studied, our ability to predict the magnitude, timing, and duration of its downstream transport is hampered by the limited predictive capabilities of existing models.

Large-scale, high temporal resolution nitrate models are limited in part due to the difficulty in simulating the dynamic nature of nitrate transport (Lucas et al., 2025; Pandit et al., 2025). Nitrate can be flushed or diluted during an event (Husic et al., 2023); it can be sourced locally or distally to the stream network (Vaughan et al., 2017); it can be generated and removed through multiple transformation pathways (Burns et al., 2016); it can be transported by surface and subsurface pathways (Miller et al., 2017); and it does not necessarily follow predictable seasonal or hydrodynamic patterns (Gorski & Zimmer, 2021). Existing modeling efforts to address these questions have been applied to smaller regional scales (e.g., the state of Iowa, Saha et al., 2023) or provide results at coarser temporal resolutions (e.g., annual or decadal, Robertson & Saad, 2021). Robust flow and nitrate models

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are necessary for constraining estimates on the source, timing, and magnitude of nitrate loading to downstream waters (Wang et al., 2024). Developing accurate models that can provide high-resolution temporal predictions and short-term forecasts can help inform on-field management, such as timing of fertilizer application (Tamagno et al., 2022) or supplemental irrigation (Sangha et al., 2023), if anticipated nitrate losses are high. In the MRB, nitrate loading is often used as a target for management, such as in the Gulf Hypoxia Action Plan (Mississippi River/Gulf of Mexico Watershed Nutrient Task Force, 2008), thus reducing uncertainty in its estimation is paramount.

While traditional approaches to predicting nitrate processes and informing management exist, they face significant hurdles. Process-based models have a strong physical basis and are highly interpretable, but they are often difficult to calibrate, are typically applicable only to the watershed where they are developed, and have limited transferability to new sites (Lucas et al., 2025). Simpler data-driven methods, like the regression-based LOADEST (Runkel et al., 2004) and WRTDS (Hirsch et al., 2010) models, can be quite robust but their application is limited to basins with long historical records and they generally cannot make predictions in unmonitored basins (Fang et al., 2024). In contrast, machine learning architectures, such as random forest (RF) models and long-short term memory (LSTM) networks, show promise for overcoming the limitations of traditional approaches (Appling et al., 2022). Results from machine learning studies have highlighted that hydrologic and water quality data sets contain more information in them than is explained by our existing process-based formulations (Nearing et al., 2021). These approaches also excel at learning complex temporal dependencies directly from data (Kratzert et al., 2019), positioning them to overcome challenges related to process complexity, model transferability to unmonitored locations, and data requirements that limit previous methods in water quality prediction.

For streamflow prediction, the accuracy of machine learning models has surpassed that of process-based models (Höge et al., 2022; Kratzert et al., 2018; Ouyang et al., 2021). In particular, LSTM networks, a class of deep learning models developed for learning temporal trends in the data (Hochreiter & Schmidhuber, 1997), have proven particularly successful for reproducing hydrologic timeseries (Lees et al., 2022) and capturing non-linear interactions (Kratzert et al., 2018). In the field of water quality modeling, continental-scale LSTMs have successfully been applied to simulate riverine solute concentrations (Fang et al., 2024), water temperature (Rahmani et al., 2021), dissolved oxygen (Zhi et al., 2023), and sediment concentrations (Gupta & Feng, 2025; Song et al., 2024). Yet despite their versatility, machine learning models are not without limitations (Zhi et al., 2024). Training machine learning models requires extensive and high-quality data, which may exclude large swaths of data-scarce regions from consideration (Fang et al., 2024). Further, if models are overfit, their transferability and generalization to unmonitored locations is challenging (Willard et al., 2025).

Machine learning models perform best using large data sets from which broad-scale patterns can be extracted (Kratzert et al., 2024). Hydrological data, in the form of streamflow observations, is often widely available. Across the contiguous US (CONUS), the US Geological Survey (USGS) monitors more than 13,500 stream gages that record continuous data. However, nitrate data are infrequently collected via discrete samples at daily, weekly, monthly, or annual intervals. This lack of intensively measured data presents a significant challenge for data-driven modeling approaches for estimating constituent loads (Saha et al., 2024). Yet the relatively recent advent and deployment of in situ nitrate sensors, which measure sub-hourly nitrate variations (Rode et al., 2016), have the potential to provide the robust and high frequency data necessary for successfully training machine learning models across the CONUS (Fang et al., 2024). In the two decades since the first nitrate sensor was installed in the U.S (Burns et al., 2019), researchers and government agencies have deployed hundreds more. The coincident rise of machine learning tools and sensor data availability (e.g., nitrate, salinity, and turbidity) provide an opportunity to overcome existing modeling limitations and provide accurate predictions of yields across large spatial extents in the US (Gorski et al., 2024; Kemper et al., 2025; Smith et al., 2024).

Understanding why a model gives the prediction that it does can be as useful to a modeler and stakeholder as the accuracy of the prediction (Husic, 2025a; Lundberg et al., 2020). To this end, model explainability methods, such as Shapley values (Shapley, 1953), have allowed researchers to peer into black-box machine learning models in order to interpret predictions (Maier et al., 2024). Such interpretability tools can reveal potential latent factors that shape riverine nitrate levels. For example, these explainable interpretations can be used to identify (1) dominant drivers of nitrate across space and (2) potential thresholds where the influence of specific catchment attributes on riverine nitrate become substantial (Husic, 2025a). For instance, understanding at what density tile drainage transitions from a minor factor to a dominant pathway for nitrate loss (Cain et al., 2022), or quantifying the extent of wetland restoration needed to achieve substantial nitrate reduction benefits (Cheng et al., 2020), remains a significant challenge. Addressing this gap in process understanding—characterizing these potentially non-linear relationships and threshold responses for key anthropogenic and natural landscape features—is essential for developing targeted and efficient management strategies to mitigate nitrate pollution.

We pursued four primary objectives in this study: (1) To develop LSTM models that predict the temporal variation of daily streamflow, nitrate concentration, and nitrate yield; (2) to develop interpretable random forest (RF) models that explain the spatial variation in long-term streamflow, nitrate concentration, and nitrate yield; (3) to identify thresholds beyond which catchment characteristics exert strong control over riverine nitrate concentrations; and (4) to estimate the proportion of surplus nitrogen exported from subbasins in the Mississippi River Basin (MRB). To address these objectives, we first trained large-scale LSTM models using CONUS-wide data to predict daily streamflow, nitrate concentrations, and nitrate yields. Next, we constructed RF models and applied Shapley value-based interpretability to determine threshold behavior of catchment attributes to drive spatial patterns in long-term water quality. We applied domain adaptation techniques to transfer the LSTM models to the MRB and generated daily predictions for 505 subbasins draining to the Gulf of Mexico. Finally, we combined modeled nitrate yield with a nitrogen surplus data set to quantify the fraction of excess nitrogen exported from the landscape by rivers. By estimating temporally dynamic nitrate export and its influential factors across space, we develop novel insights to help decision-makers who require finer temporal-resolution estimates, such as for event-based versus long-term nutrient management questions.

## 2. Materials and Methods

### 2.1. Model Inputs

Stream nitrate and flow data were retrieved from the Water Data for the Nation online repository (Figure 1; USGS, 2023). A total of 155 stations in CONUS have high-frequency nitrate sensor data available during our period of interest. However, many of these sensors are in locations other than streams, such as lakes, coasts, canals, karst springs, and bioreactors. We excluded all non-riverine sites, leaving 95 sensor sites for our analysis, with a data range from 2007 to 2022. The amount of sensor data over this period varied, from as little as a few months of data at some sites to more than a decade at others (Figure S1 in Supporting Information S1). In total, 158,282 daily averaged observations of nitrate were derived from the sensors, corresponding to 433 years of data. The median amount of sensor data per site was approximately 3.5 years (Figure S1 in Supporting Information S1). We retrieved flow data from each nitrate sensor site; however, some sensors were not co-located with stream gages in which case we retrieved flow data from the nearest gage up to 20 km away along the same river (Cai et al., 2021). To increase the amount of data for streamflow model training, we added 659 sites from the Catchment Attributes and Meteorology for Large-sample Studies (CAMELS) to our analysis (Addor et al., 2017), bringing the total number of flow sites to 754, with a data range from 2000 to 2022. The inclusion of CAMELS sites provides good spatial coverage and a diverse range of stream conditions, which tends to improve overall global training of LSTM models (Kratzert et al., 2024). For the 95 sites where both flow (mm/d) and nitrate concentration (mg-N/L) were available, we calculated nitrate yield (kg-N/ha/day) as the area-normalized product of flow and concentration. The distributions of mean flow, nitrate concentration, and nitrate yield for observation sites are shown in Figure S2 of Supporting Information S1.

The model forcings included dynamic and static variables. We retrieved daily meteorological forcings for seven dynamic variables (precipitation, shortwave radiation, minimum and maximum temperature, hours of daylight, and snow water equivalent) from the Daymet data set (Thornton et al., 2014). Daymet provides daily gridded estimates (1 $km^2$) of meteorological forcings, corrected using ground observations. Using basin boundaries for each sensor location, derived from PyNHD (Chegini et al., 2021), we calculated the daily area-weighted meteorological forcings at each site. Regarding the static catchment variables, we considered 31 catchment attributes, which we aggregated based on their likelihood to impact streamflow and nitrate processes. Of the 31 attributes, 27 were derived from the BasinATLAS data set (Linke et al., 2019), which is a global collection of 281 attributes classified under 6 categories of hydrology, physiography, climate, landcover, soils and geology, and anthropogenic. The remaining 4 attributes were atmospheric nitrate deposition, farmland fertilizer application, non-farm fertilizer application, and tile drainage density. Long-term records of atmospheric deposition of nitrate were obtained from National Atmospheric Deposition Program (NADP, 2019). Spatial deposition estimates were created by applying inverse distance weighting to the point measurements made at NADP sites. For fertilizer

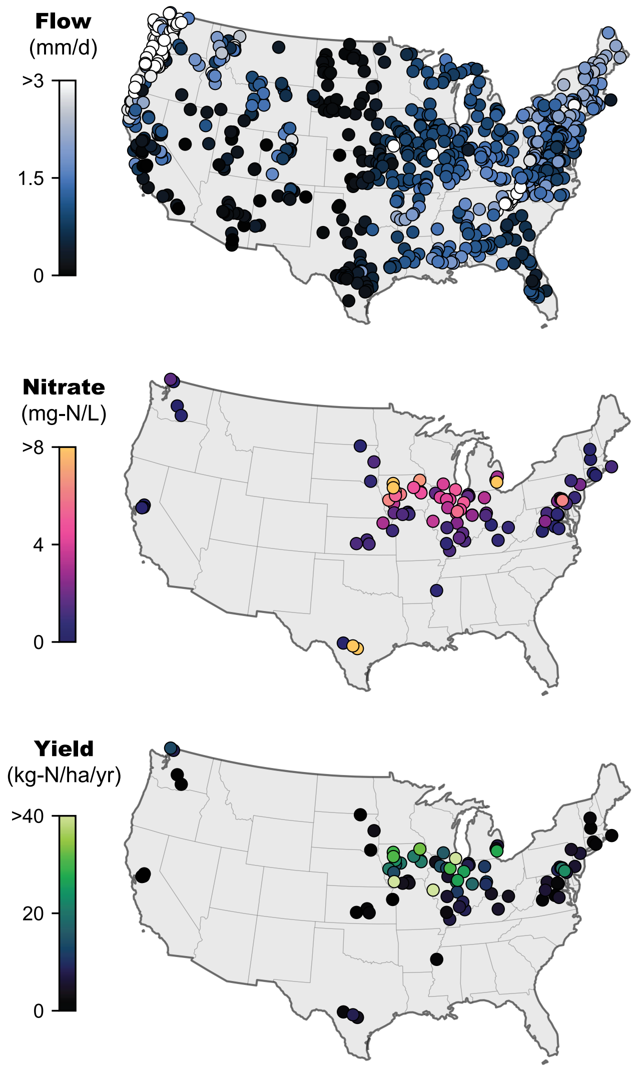


**Figure 1.** Mean observed streamflow ($n$ = 754), nitrate concentration ($n$ = 95), and nitrate yield ($n$ = 95) at sites used for model training and evaluation.

application, we used estimates of farm and nonfarm nitrogen application (Falcone, 2021). Although this fertilizer data set contains annual estimates (1987–2012), it does not cover the full duration of our study (1980–2022). Thus, we use the long-term average for both fertilizer application types to avoid extrapolating outside of the data range. Tile-drainage density for each basin was estimated from the AgTile-US data set (Valayamkunnath et al., 2020). All catchment attributes, their description, and their source (Table S1 in Supporting Information S1) and a correlation matrix describing feature collinearity (Figure S3 in Supporting Information S1) are listed in the supplemental material.

### 2.2. Model Development

We trained two large-scale LSTM models, one for streamflow (mm/d) and one for nitrate concentration (mg-N/L), using the dynamic and static inputs described above. The estimated annual nitrate load (kg-N/yr) was calculated as the product of the flow and concentration LSTM model outputs. We facilitated comparison across watersheds of different sizes by normalizing the nitrate load by drainage area to calculate the nitrate yield (kg-N/yr/ha). The advantage of training two models (streamflow and nitrate concentration separately) instead of a single model (nitrate yield directly) was that it allowed us to assess prediction accuracy of each component independently.

The LSTM modeling architecture includes recurrent neural networks with the capability of learning long-term dependencies while adapting to short-term variations in the data (Hochreiter & Schmidhuber, 1997). The framework uses gates (input, forget, and output) to regulate the flow of information through the network and overcome the vanishing gradient problem. We used the HydroDL codebase for our LSTM application (Feng et al., 2020), which is a PyTorch-based code tailored for hydrologic modeling. The inputs to the two HydroDL models were nearly identical and consisted of 7 meteorological forcings and 31 static catchment attributes. The nitrate concentration model had one additional dynamic input, which was either the observed streamflow (when available) or the modeled streamflow (using the flow LSTM model to gap-fill). All inputs were pre-processed using Z-score normalization prior to model training. The training target for the streamflow LSTM was daily-averaged streamflow observations ($n$ = 754 sites), whereas the training target for the nitrate LSTM was the daily-averaged nitrate sensor readings ($n$ = 95 sites).

### 2.3. Model Evaluation

The models were trained globally—that is, observations from all sites were concatenated into a single training dataset—using the root mean squared error loss function and the Adadelta optimizer (Zeiler, 2012). Key hyperparameters for an LSTM include hidden state size, sequence length, dropout rate, and batch size. Fine tuning of the hyperparameters was conducted to optimize model performance. We trained LSTM models on combinations of hyperparameter sets by varying batch size from 128 to 4,096, dropout from 0.2 to 0.6, hidden size from 64 to 256, and sequence length from 30 to 365 days. The optimum set of hyperparameters was selected by assessing the convergence and the value of the loss function. For the nitrate model, the optimum hyperparameter set consisted of 2,000 training epochs, 256 hidden states, a sequence length of 180 days, a dropout rate of 0.3, and a batch size of 128 for spatial validation (to aid in generalization) and 4,096 for temporal validation (to aid in convergence). The streamflow model had the same dropout rate and hidden states but fewer epochs (500) and the same batch size (128) for both temporal and spatial validation. The models were trained in under 6 hr using NVIDIA A100 GPUs on the University of Kansas High Performance Computing Cluster and Virginia Tech Advanced Research Computing Cluster.

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Temporal and spatial cross-validation (CV) were conducted to assess the respective ability of the LSTM models to (1) forecast unseen conditions at sites included in training and (2) make predictions at (pseudo)-unmonitored basins not included in training (Kratzert et al., 2019). Both procedures used nearly identical network architecture and hyper-parameters; the training subsets were the main difference, allowing direct comparison of temporal versus spatial transferability. For the temporal validation, streamflow and nitrate concentration models were trained using five-fold CV with 80% of each fold used for training and the remaining 20% used for testing (Gharib & Davies, 2021). Each of the folds used a different part (20%) of the time series at each site for testing whereas the remaining four parts (80%) were used for training. Thereafter, the full temporal validation time series was constructed by concatenating each of the five 20% testing folds. For the spatial validation, streamflow and nitrate concentration models were trained using twelve-fold CV following the approach of Kratzert et al. (2019). As the goal of spatial CV is to assess predictions in unmonitored basins, we excluded ~8.3% of the sites in each fold entirely from training and use their full time series for validation. Lastly, when the LSTM models were transferred to make predictions for subbasins in the MRB, an ensemble of the 12-fold spatial CV model was used.

Model predictions were evaluated using three statistical metrics: Kling Gupta Efficiency (KGE; Gupta et al., 2009), Nash Sutcliffe Efficiency (NSE; Nash & Sutcliffe, 1970) and square of the correlation coefficient ($R^2$; Pearson, 1901). KGE and NSE range from $-\infty$ to 1 while $R^2$ ranges from 0 to 1. The perfect model would have a KGE, NSE, and $R^2$ of 1, with lower values indicating sub-optimal model performance. We adopted the suggested performance thresholds from Pandit et al. (2025), stating the thresholds for a "satisfactory" model of daily streamflow and nitrate simulation are 0.50 and 0.35, respectively, for NSE and 0.60 and 0.45, respectively, for $R^2$. At present, no commonly accepted performance thresholds for KGE exist. KGE is calculated by comparing the correlation, bias, and variability of observed and simulated time series; NSE measures the relative magnitude of residual variance compared to observed data variance; and $R^2$ evaluates the proportion of observed data variance explained by the model. KGE is calculated as:

$$\text{KGE} = 1 - \sqrt{(r-1)^2 + (1-\alpha)^2 + (1-\beta)^2} \tag{1}$$

where $r$ is the correlation coefficient, and $\alpha$ and $\beta$ are the ratios of the standard deviation and mean, respectively, of model predictions to data observations. NSE and $R^2$ are calculated as:

$$\text{NSE} = 1 - \left[\frac{\sum_{i=1}^{n}\left(y_{\text{obs}} - y_{\text{sim}}\right)^2}{\sum_{i=1}^{n}\left(y_{\text{obs}} - \overline{y_{\text{obs}}}\right)^2}\right] \tag{2}$$

$$R^2 = \left[\frac{\sum_{i=1}^{n}\left(y_{\text{obs}} - \overline{y_{\text{obs}}}\right)\left(y_{\text{sim}} - \overline{y_{\text{sim}}}\right)}{\sum_{i=1}^{n}\left(y_{\text{obs}} - \overline{y_{\text{obs}}}\right)^2 \sum_{i=1}^{n}\left(y_{\text{sim}} - \overline{y_{\text{sim}}}\right)^2}\right]^2 \tag{3}$$

where $y_{\text{obs}}$ are observed data and $y_{\text{sim}}$ are model predictions. A bar above either variable denotes averaging. The values of KGE, NSE, and $R^2$ reported in the text are the ensemble means that arise from temporal and spatial CV procedures. This study is particularly interested in nitrate dynamics in the MRB; thus, we report two sets of summary statistics: (1) estimates for sites in the MRB ($n = 50$ sensors) and (2) estimates for all sites in CONUS ($n = 95$ sensors).

### 2.4. Explaining Spatial Variation and Feature Thresholds

Whereas the LSTM model makes temporal predictions at sites, we developed a complementary random forest (RF) model—distinct from the LSTM model—to explain long-term mean spatial variations across sites. While methods for explaining LSTM models exist (e.g., DeepSHAP), they are at a nascent phase (Lee et al., 2022; Zhi et al., 2024) and scale poorly with a large number of observations and features (Van den Broeck et al., 2022). Thus, we do not explain the internal workings of the LSTM nitrate models in this work, as others have noted that their predictions are largely driven by temporal flow behavior (Gorski et al., 2024). Rather, we explain spatial

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variations in long-term mean flow, nitrate concentration, and nitrate yield dynamics using the RF model, which has a more readily interpretable modeling architecture.

The RF models consisted of 1,000 regression trees and each was trained to predict the observed long-term means of flow, nitrate concentration, and nitrate yield at the 95 sensor sites. The predictors (features) used in the RF models were the 31 static catchment attributes also used in LSTM training (Table S1 in Supporting Information S1). Each tree was trained on an in-the-bag bootstrap of observations and validated on the out-of-bag observations (Li et al., 2010). The out-of-bag model had strong predictive performance ($R^2 \geq 0.57$ for all three model targets; Table S2 in Supporting Information S1), providing us with confidence in explaining the model. Further, a comparison of long-term mean nitrate concentration predicted by the RF ($N_{\text{RF}}$) to that of the LSTM ($N_{\text{LSTM}}$) showed close alignment ($N_{\text{LSTM}} = 1.08 \times N_{\text{RF}}$, $R^2 = 0.81$).

To explain the RF model, we calculated Shapley values for the long-term mean flow, concentration, and yield predictions. Shapley values ($\psi$) are a concept from cooperative game theory that explain how much an input feature contributes to a model prediction (Lundberg & Lee, 2017; Shapley, 1953). We used the TreeSHAP algorithm with an interventional value function (Lundberg et al., 2020). This approach assesses the following question: "how would the prediction at a site change if one could go in and intervene to change the value of a feature to something other than its true value?". For example, given a watershed in Iowa with 60% tile drainage and 5% wetland cover, this approach could assess the questions: what would the predicted nitrate concentration at this site be if (1) the tile drain density were reduced from 60% to 0% or if (2) the wetland cover were doubled from 5% to 10%? Shapley values can also reveal feature threshold behavior, that is, points beyond which a feature begins to exert considerable, potentially non-linear influence on river chemistry. Further details on the strengths and caveats of applying Shapley values to research questions in catchment science can be found in Husic (2025a).

### 2.5. Model Application in Mississippi River Basin

Using transfer learning, our spatially validated flow and nitrate concentration LSTM models were applied to applicable basins in the MRB to estimate nitrate yields. The USGS Hydrologic Unit Code 8 (HUC-8) basins were selected as the spatial unit for transferring our models because the median basin area in our training data set (3,443 $\text{km}^2$) was similar to the median MRB HUC-8 drainage area (3,450 $\text{km}^2$). Predictions from machine learning models become unstable when new inputs substantially differ from the range of inputs that the model was trained on (Khoshkalam et al., 2025). In other words, machine learning models interpolate well but extrapolate poorly. Thus, while there are a total of 818 HUC-8s in the MRB, only a subset have similar catchment attributes as the sites used in LSTM model training. To identify this subset, we used transformation-based domain adaptation techniques to maintain consistency in model transfer (Farahani et al., 2021). To evaluate the similarity of trained nitrate sites ($n = 95$) to potential transfer basins ($n = 818$), we used a combination of two techniques: *t*-SNE (*t*-distributed Stochastic Neighbor Embedding) and DBSCAN (Density Based Spatial Clustering Application with Noise), which are explained below.

*t*-SNE is a nonlinear dimensionality reduction technique (van der Maaten & Hinton, 2008), which we use to downscale the catchment attributes from the complete 31-dimensional space into a 2-dimensional embedding. The aim of *t*-SNE is to preserve the local proximity of points. Simply put, basins close to one another in the 31-D space should be close to one another in the 2-D embedding. The resulting 2-D embedding of the catchment attributes for training sites and potential HUC-8 transfer basins was clustered, using DBSCAN, to generate two dominant clusters: (1) Transfer basins similar to the training sites and (2) transfer basins dissimilar to the training sites. DBSCAN identifies core points in the 2-D space, which contain a pre-defined minimum number of points that are within a prescribed distance from the core point (Ester et al., 1996). The algorithm iteratively forms clusters by including all points within a set distance of a core point and continues to expand the cluster by including additional neighbors. This process is repeated until the density-connected region is fully explored and points that do not meet these criteria are classified as noise.

Of the 818 HUC-8 basins that constitute the MRB extent upstream of Baton Rouge, Louisiana, our domain adaptation approach identified 505 as "similar" to sites used in training (Figure 2). A significant spatial pattern emerged where basins east of the 98th meridian were classified as similar, whereas basins west of this meridian were classified as dissimilar. This result is reasonable as most of the nitrate sites used in model training fell east of the Great Plains (Figure 1) and thus encode for behavior that is captured by the catchment attributes at those

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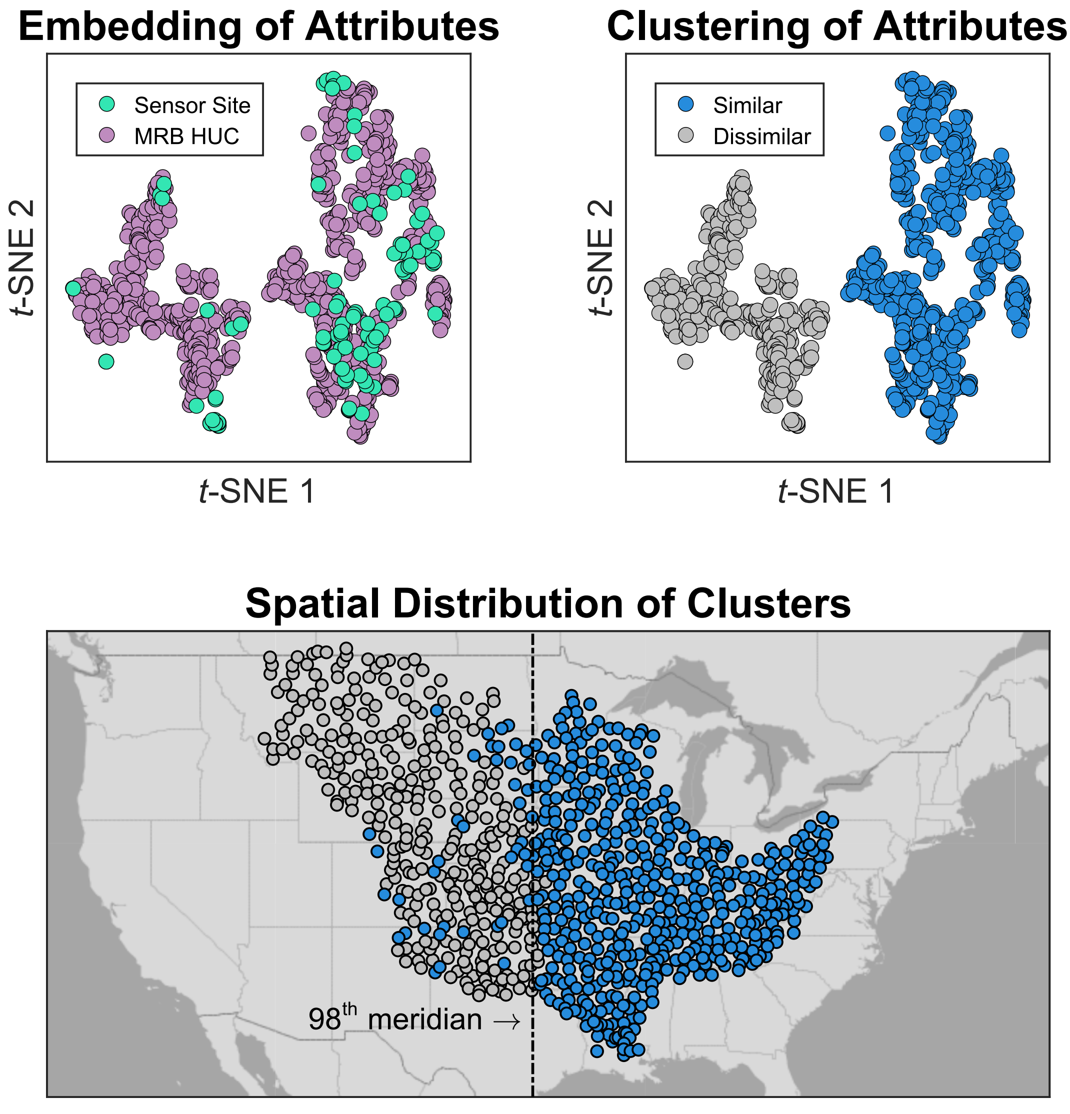


**Figure 2.** Domain adaptation procedure for transfer learning of model to the MRB. *t*-SNE plot of the 2-D embedding for the original 31-D catchment attribute space. The dimensinality reduction analysis shows that most sensor sites (83 of 95) are located in the right cluster along with most HUC (505 of 818) basins. HUC basins in the same cluster as the training sites are labeled "Similar" whereas HUC basins in the opposite cluster are "dissimilar." Spatially, a bifurcation appears around the 98th meridian, and we apply our models to basins east of this meridian. *t*-SNE = *t*-distributed stochastic neighbor embedding.

locations. Previous work in the MRB indicated that the lands east of the 98th meridian account for more than 85% of the nitrate delivery to the Gulf of Mexico (Crawford et al., 2019). Hence, while our model applies to approximately 60% of the total area of the MRB, the regions where it applies are the most problematic with regard to nitrate pollution. Hereafter, all MRB results and references apply to these 505 basins rather than the full 818 basins.

Upon the successful transfer of our spatially validated model, we applied it to generate daily streamflow, nitrate concentration, and nitrate yield estimates for 505 HUC-8 basins in the MRB, starting in 1980 and ending in 2022. These HUC-8 model estimates are incremental in that they only reflect the water and nitrate produced exclusively by drainage area contained within the HUC-8 basin (i.e., there is no consideration of routing of material through and across other basins). Model inputs for the HUC-8 basins, such as meteorological forcings and catchment attributes, were retrieved in the same manner as for the nitrate sensor sites.

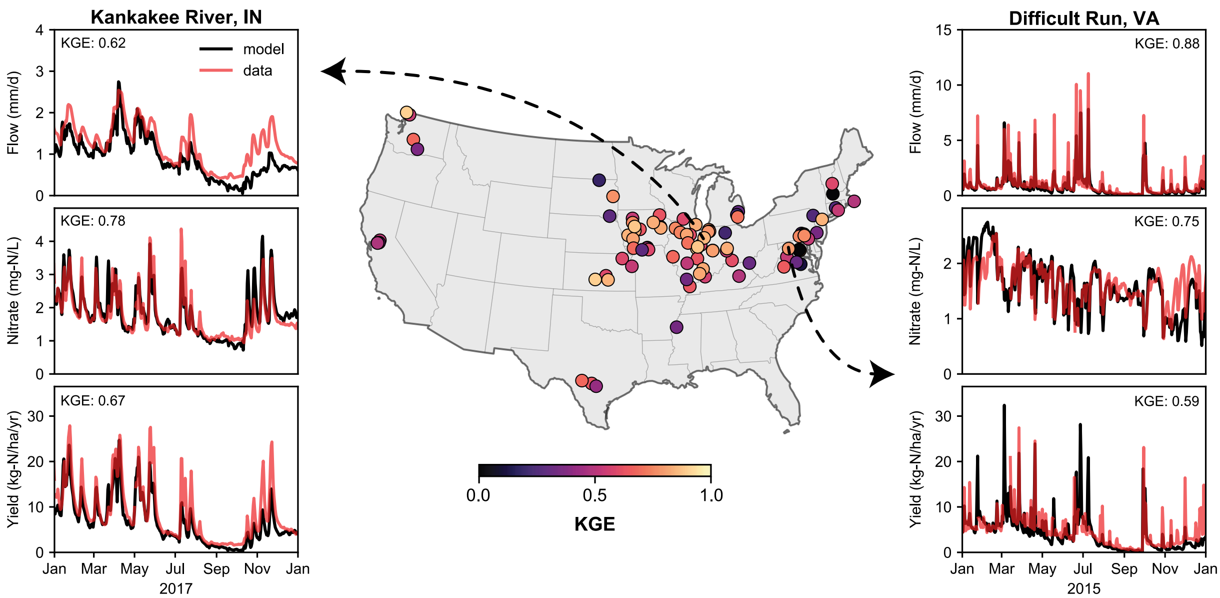


**Figure 3.** Temporal cross-validation model performance for watersheds in CONUS ($n = 95$). A 1-year snapshot of the modeled flow, concentration, and yield timeseries for two sites is shown: Kankakeee River, IN (USGS 05515500), and Difficult Run, VA (USGS 01646000). The model captures the differing hydrologic characteristics and nitrate responses to storm events at the two sites. Difficult Run (right) shows influence of nitrate dilution while the Kankakee River (left) shows nitrate mobilization during events. KGE, Kling-Gupta Efficiency.

To estimate the proportion of surplus nitrogen exported as nitrate from each HUC-8, we defined a metric termed Nitrogen Delivery Ratio (NDR), which represents the fraction of the excess nitrogen in a system that is hydrologically exported. The NDR is the complement to another common metric termed nutrient retention (R; Stackpoole et al., 2021), that is NDR = 1−R. The NDR metric was calculated as

$$\mathrm{NDR} = \frac{N_{\mathrm{export}}}{N_{\mathrm{surplus}}} \tag{4}$$

where $N_{\mathrm{export}}$ is the annual nitrate exported (kg-N/ha/yr), as estimated by our yield model, and $N_{\mathrm{surplus}}$ is the annual nitrogen surplus (kg-N/ha/yr) in a basin, as estimated by the Trajectories Nutrient Data set for Nitrogen (TREND; Byrnes et al., 2020). TREND provides annual county-level $N$ mass balances for 88 years (1930–2017). This mass balance includes inputs (atmospheric deposition, biological $N$ fixation, fertilizer and manure applications, and human waste) and an output (crop uptake). The net balance of these fluxes, spatially averaged over a basin, results in the net nitrogen surplus available for transport.

## 3. Results

### 3.1. Observed Flow and Nitrate Variability

Mean flow ranged from a low of 0.002 mm/d in Bonita Creek, AZ, to a high of 9.2 mm/d in the Hoh River, Washington (Figure 1). Mean nitrate concentrations ranged from a low of 0.17 mg-N/L at Sacramento River near Freeport, California, to a high 14.91 mg-N/L at San Antonio River near Elmendorf, Texas. The highest nitrate yields generally occurred in the Midwest. The greatest mean nitrate yield was 62.7 kg-N/ha/yr in the Illinois River at Florence, Illinois, and the lowest was 0.18 kg-N/ha/yr at Cache Stream in Rio Vista, California.

### 3.2. Deep Learning Predictions in Monitored Basins

The geographic distribution of the nitrate concentration model metrics for the temporal CV model revealed the strongest model performance in the Midwestern US (Figure 3). The highest performance for nitrate concentration model (KGE ≈ 0.90) was achieved at Raccoon River, Iowa (USGS 05484500) and Spoon River, Illinois (USGS 03336890). Performance at these sites is aided by long sensor records (9.6 years at Raccoon River and 7.1 years at Spoon River) as well as high mean nitrate concentrations (6.53 mg-N/L at Raccoon River and 4.22 mg-N/L at

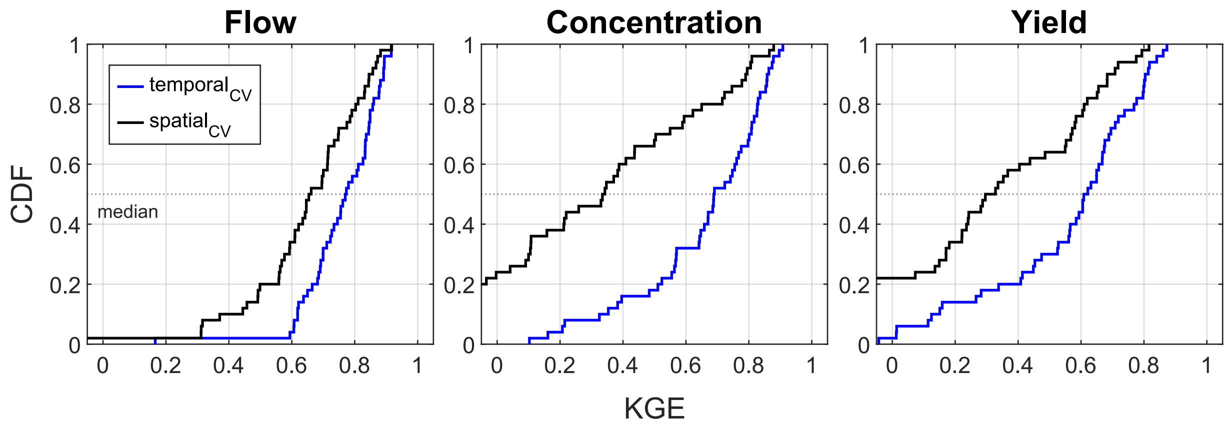


**Figure 4.** Accuracy comparison of flow, nitrate concentration, and nitrate yield models as evaluated by the Kling-Gupta Efficiency for sensor sites in the Mississippi River Basin (MRB, $n = 50$). Model results are shown for temporal (blue line) and spatial (black line) cross-validation (CV). Temporal CV excludes portions of time at each gage from training, whereas spatial CV excludes entire watersheds from training (i.e., prediction in ungaged basins). Nitrate yield estimates are the product of the flow and concentration models. Figure S4 in Supporting Information S1 contains a version of this figure for all CONUS sites ($n = 95$).

Spoon River). High mean nitrate concentrations tend to be accompanied by anthropogenic variability that is well-described by the catchment attribute data set, for example, crop cover and tile drain density. On the other hand, the site with the lowest performance (KGE = −2.62) was The Glade at Reston, Virginia (USGS 0164579522). This site is the second smallest basin in the data set (2.95 $km^2$), had low mean nitrate concentration (0.60 mg-N/L), and only had 1.6 years of data. The variability in nitrate at sites with low concentrations may be driven by latent drivers that are not captured by our catchment attribute data set, such as in-stream uptake. Further, the small size of the basin may indicate that the continental-scale dynamic and static inputs used to drive the model do not resolve local-scale conditions. Other underperforming sites, like the Connecticut River, Massachusetts (KGE = −0.25, USGS 01161280), had low mean nitrate concentration (0.21 mg-N/L), a short data record (1.1 years), and extensive upstream flow storage by dams, thus non-natural hydrologic processes control flow and nitrate delivery to the river. A third site with poor performance was Muddy Creek, Pennsylvania (KGE = −0.11, USGS 01577500). This site had a moderate data record of 2.1 years and a mean nitrate concentration of 6.14 mg-N/L, thus we anticipated the model would perform well. While the model accurately represents the mean concentration and variability of nitrate at this site, the timing of modeled flushing and dilution were frequently out of phase with observations at this site.

In general, temporal predictions of the flow, concentration, and yield models were able to capture differing hydrologic and nitrate regimes across CONUS (Figure 3). The timeseries predictions for two sites, the Kankakee River, Indiana (USGS 05515500), and Difficult Run, Virginia (USGS 01646000), demonstrate model robustness for capturing distinct patterns in river response. For the Kankakee River, stream discharge is relatively stable with long baseflow recessions and subtle flow peaks. At this agricultural site, nitrate is mobilized during storm events, with peaks in concentration generally coinciding with peaks in flow. On the other hand, flow in Difficult Run is extremely flashy, with peak discharge values that are several orders of magnitude greater than baseflow. At urbanized Difficult Run, nitrate is frequently diluted in response to storm events as wastewater effluent maintains high baseflow concentrations. The model successfully captures both the flashy discharge response and the nitrate dilution at Difficult Run. The ability of the models to capture mobilization, dilution, and seasonal responses highlights the strength of LSTM modeling to capture signals at multiple temporal scales.

### 3.3. Deep Learning Predictions in Unmonitored Basins

For the 50 study sites in the MRB, model skill was consistently higher when time periods at known gages were withheld (temporal CV) than when entire watersheds were excluded from training (spatial CV, or prediction in unmonitored basins; Figure 4). For streamflow, the median KGE dropped modestly from 0.77 under temporal CV to 0.66 under spatial CV, indicating the flow LSTM was able to generalize reasonably well. For nitrate concentration, the drop in skill was greater as median KGE declined from 0.69 to 0.34, indicating that the nitrate model performs accurately if informed by local training data, but it struggles more with generalizing in space. The nitrate yield skill followed a similar pattern as concentration, from a median KGE of 0.62 to 0.31, reflecting the combined error sources from flow and concentration models. For all 95 sites in CONUS (Figure S4 in Supporting Information S1), the temporal-to-spatial drop in median KGE was similar for flow (0.78–0.65) and nitrate yield (0.59–0.32), but much larger for nitrate concentration (0.60–0.18), further echoing that the nitrate model can perform well if informed by local data, but that it struggles to generalize to new locations, particularly those outside of where a bulk of training sensor data resides (i.e., sensors in the Midwest).

We inspected the three components that constitute the KGE metric ($r$, $\alpha$, and $\beta$) to identify how the temporally validated flow and nitrate LSTMs differ in their underlying modeled representation of the observed data (Figure S4 in Supporting Information S1). We describe the variability of the three components using the 2.5th, 50th, and 97.5th percentiles derived from their distributions. The linear correlation ($r$) between simulated and observed time series for the flow model is 0.84 [0.17, 0.94] compared to 0.69 [−0.10, 0.92] for the nitrate model, indicating that the flow model predictions more closely approximate the timing and shape of the observations than do the nitrate model predictions. The variability ($\alpha$, ratio of modeled to observed series standard deviations) for flow is 0.86 [0.29, 1.49] and for nitrate it is 0.85 [0.48, 1.14], indicating comparable under-estimation of variability by both models. The mean bias ($\beta$, ratio of modeled to observed series means) for flow is 1.04 [0.70, 3.38] and for nitrate it is 0.97 [0.77, 1.25] showing that mean flow predictions at some sites were considerably higher than observations, whereas the nitrate model had very little bias. In addition to KGE, the statistics for the NSE and $R^2$ of temporal and spatial CV models are shown in Figures S5 and S6 of Supporting Information S1.

The hydrologic and water quality modeling community has not yet established recommended model performance metrics for KGE, thus we use NSE criteria from Pandit et al. (2025) and others to contextualize our large-scale flow, concentration, and yield model predictions. Starting with temporal CV of daily flow, 45% of our sites qualify as having very good performance (NSE $\geq$ 0.80), 29% as good (NSE $\geq$ 0.70), 24% as satisfactory (NSE $\geq$ 0.50), and 2% as limited (NSE < 0.50). For the spatial CV model of daily flow, these values are 17%, 34%, 32%, and 18%, respectively. For the temporal CV model of daily nitrate concentration, 36% of our sites qualify as having very good performance (NSE $\geq$ 0.65), 27% as good (NSE $\geq$ 0.50), 16% as satisfactory (NSE $\geq$ 0.35), and 21% as limited (NSE < 0.35). For the spatial CV model of daily nitrate for all sites ($n = 95$), these values are 15%, 1%, 2%, and 82%, respectively, whereas for MRB sites ($n = 50$), the values 24%, 6%, 2%, and 68%, respectively.

Based on NSE guidance, a considerable fraction of spatially validated sites have "limited" performance for two reasons (1) the NSE metric punishes missed timing even if overall mean and variability behavior is well-represented at a site and (2) deep learning models of water quality (or hydrology) are not necessarily entity aware, that is, catchment attributes may help improve predictions at gaged locations without necessarily learning the underlying relationships needed to generalize to new catchments (Heudorfer et al., 2025). To the first point, for nitrate concentration, where the observed series has relatively low variance compared to flow, even modest absolute errors (caused by a slight mismatch in the timing and shape of modeled predictions) can push the NSE to very negative values (Gupta et al., 2009). If we instead consider the nitrate yield modeled estimates, which partly compensate for this issue through multiplication of concentration by flow, the proportion of sites having very good, good, satisfactory, and limited performance for spatial CV of nitrate yield is 15%, 19%, 16%, and 51%, respectively, for all sites ($n = 95$) and 24%, 14%, 16%, and 46%, respectively, for MRB sites ($n = 50$). While the fraction of "limited" sites is still considerable, nitrate yield estimates are more accurate as they are compensated by a strong flow model.

### 3.4. Drivers of Spatial Variation in Nitrate

The RF model performed well for the prediction of spatial variations in long-term mean flow ($R^2 = 0.81$), nitrate concentration ($R^2 = 0.64$), and nitrate yield ($R^2 = 0.57$) (Table S2 in Supporting Information S1). The Shapley values ($\psi$) for the long-term flow model unsurprisingly indicate that precipitation, aridity, and soil water content are the most influential (Figure 5). The deposition of nitrate is also indicated as a lesser influential variable, which is likely due to its correlation with precipitation for the sites in our analysis ($n = 95$, Figure 1). This underscores that while interpretable techniques can reveal how machine learning models transform inputs to outputs, these interpretations are not necessarily causal and should be interrogated critically (Husic, 2025a).

Whereas mean flow was dominated by a few drivers, mean nitrate concentration and yield estimates were influenced by the complex interaction of many drivers (Figure 5), as shown in prior studies (Sadayappan et al., 2022; Wherry et al., 2021). Well-known drivers appear as the five most influential predictors for nitrate

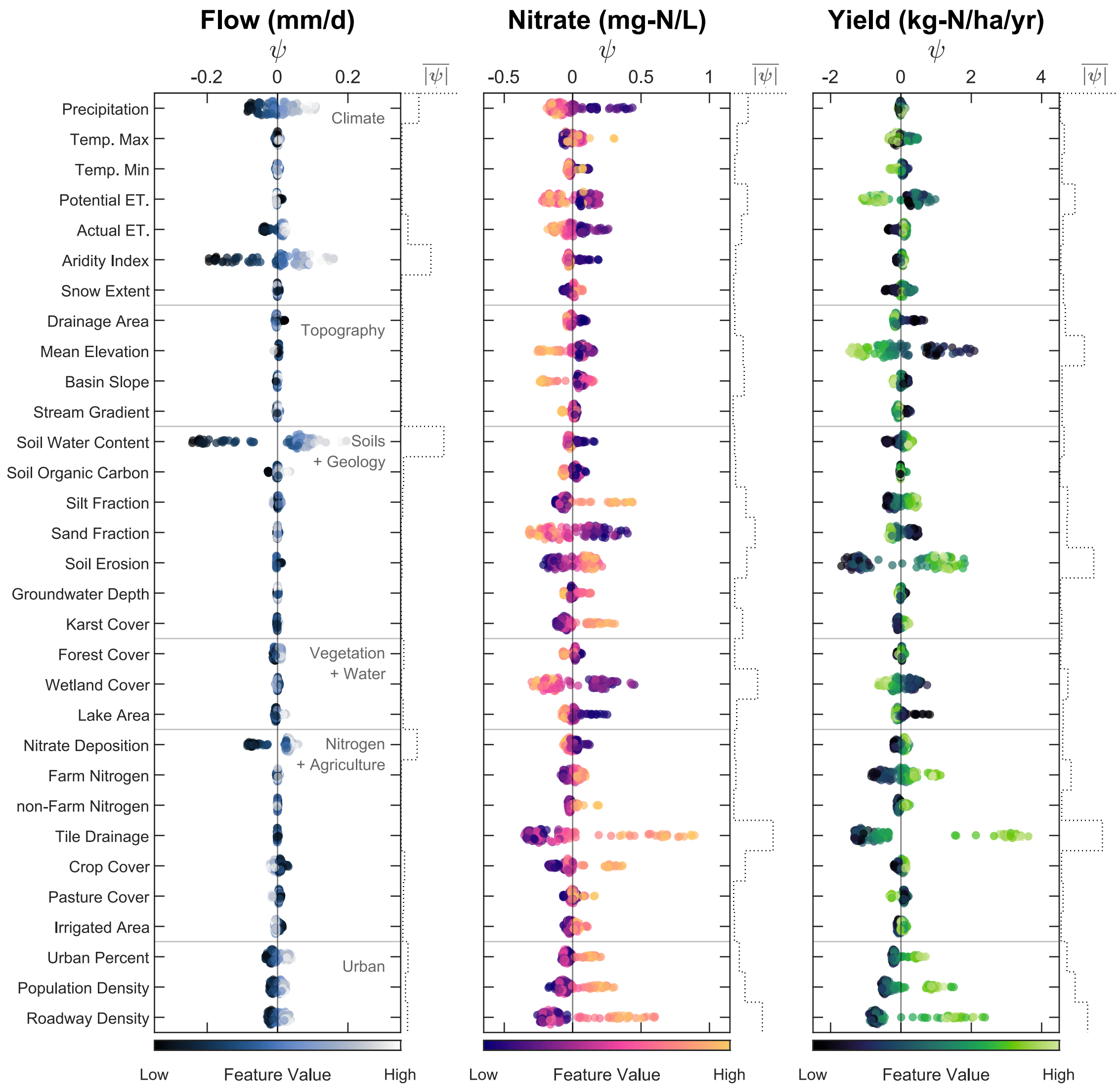


**Figure 5.** Shapley value plots summarizing how individual watershed attributes influence random-forest predictions of long-term mean flow, nitrate concentration, and nitrate yield at sites with nitrate sensors ($n = 95$). Each point represents a single basin; its horizontal position is the Shapley value ($\psi$), showing how much that feature pushes the prediction above ($\psi > 0$) or below ($\psi < 0$) the basin-average baseline. Point color encodes the relatively magntiude of the predictor in that basin (dark = low, light = high). The dotted-line to the right of each plot indicates the absolute importance of a feature to the model. Horziontal brackets group variables into thematic classes.

concentration prediction, ranked from largest to smallest $\overline{|\psi|}$, including (1) tile drainage, (2) roadway density, (3) wetland cover, (4) sand fraction, and (5) precipitation. For nitrate yield, the five most influential drivers include (1) tile drainage, (2) soil erosion, (3) roadway density, (4) elevation, and (5) population density. As the nitrate yield estimates are a product of flow and concentration drivers, there are instances where a predictor can be positively related to flow but negatively related to concentration, thus the predictor's influence on nitrate yield is effectively canceled out. An example is precipitation, where high precipitation leads to greater flows but diluted nitrate concentrations. The result is that long-term mean precipitation has little to no influence on explaining the spatial variation in nitrate yield (although interannual variations in precipitation are important as explored in Section 3.5). Conversely, the ability of a predictor to explain spatial variation can be magnified if its influence on flow and concentration is constructive, such as in the case of the urban predictors, where roadway density compounds increased flow with increased nitrate concentrations. A third case is where a predictor heavily influences nitrate but has little influence on flow, such as the case for wetland cover. The result is that wetlands were very important for explaining variation of nitrate concentration but less so for explaining variation in yields.

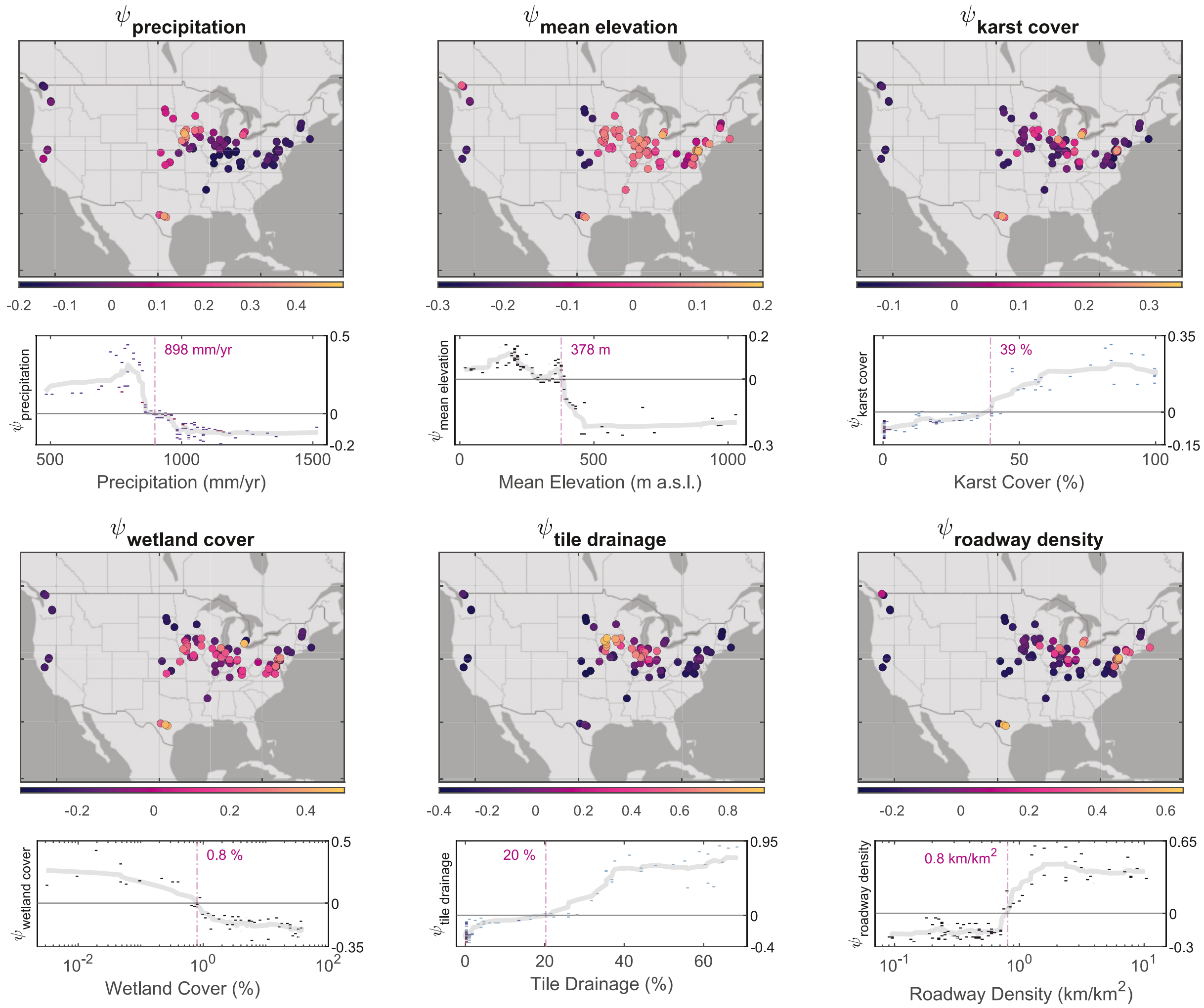


**Figure 6.** The spatial distribution and thresholds of nitrate concentration drivers for selected influential features, one from each of the six thematic groupings (see Figure 5). The map and partial dependence plots show the Shapley values ($\psi$) for selected features, which has units of mg-N/L. The relative importance of a feature to increase ($-\psi$) or decrease ($+\psi$) the random forest model prediction is indicated by the magnitude of $\psi$. The partial dependence plot of each feature is shown with features values for individual basins represented by markers (−). The point at which $\psi$ crosses $\psi = 0$ indicates a potential threshold, which is identified by a vertical dash-dot line and text indicating the abscissa value. A moving average of feature values is indicated by a gray line to show general trends. Figure S7 in Supporting Information S1 contains a version of this figure for identifying nitrate yield thresholds.

Thresholds beyond which catchment attributes begin to substantially influence nitrate concentrations are shown in Figure 6. The marginal contribution of precipitation ($\psi_{\text{precipitation}}$) to nitrate concentration prediction goes from positive to negative at approximately 900 mm/yr. Below this threshold, lower precipitation contributes to greater ($+\psi$) modeled nitrate whereas higher precipitation contributes to lower ($-\psi$) modeled nitrate. For mean elevation, the threshold of $\psi_{\text{mean elevation}}$ occurs at 378 m a.s.l., with low lying elevations contributing to higher ($+\psi$) nitrate predictions. The extent of karst or carbonate cover in a basin does not substantially influence nitrate concentrations until a threshold of around 40%, at which case the presence of karst increases ($+\psi$) modeled predictions. For wetland cover, the lack of wetlands (<0.8%) results in higher modeled nitrate ($+\psi$) compared to the presence of wetlands (>0.8%), which results in reduced modeled nitrate ($-\psi$). A non-linear threshold is also identified for tile drainage, where a threshold of 20% is identified, beyond which tile drainage considerably increases ($+\psi$) modeled nitrate. For roadway density, a threshold of approximately 0.8 km of roadway per 1 km$^2$ of drainage was identified. It is important to note the context in which these thresholds are identified and how they should be interpreted. These results do not necessarily mean that, for example, tile drainage values below 20% do not influence nitrate concentrations, but rather that for the predominantly agricultural sites considered in our analysis, this was the threshold that helped to distinguish tile drainage influence across sites. For catchment attributes that can be modified, such as wetland cover, tile drainage, and roadway density, the identification of thresholds can inform management targets and conservation strategies.

### 3.5. Predictions in the Mississippi River Basin

Daily flow, nitrate concentration, and nitrate yield predictions were generated for 505 basins in the MRB from 1980 to 2022 using the spatially validated models (Figure 7). Model results hereinafter are reported as the mean ± one standard deviation of all 505 HUC basins or, when denoted, a subset of all basins (e.g., those in Iowa). Average annual flow for the MRB basins over this period was 0.83 ± 0.40 mm/d and the mean nitrate concentration was 1.57 ± 1.19 mg-N/L. Modeled average nitrate concentrations ranged from a high of 7.61 mg-N/L in the Nishnabotna Basin (HUC-8 ID: 10240004), which straddles Iowa, Nebraska, and Missouri, to a low of 0.26 mg-N/L in the Lower Canadian-Walnut Basin, OK (11090202). Eight of the top 10 long-term mean nitrate concentrations were from HUC-8 basins draining parts of Iowa. The EPA standard for nitrate (10 mg-N/L) was exceeded at least once (over the 42-year study period) at 87 of the 505 basins. This standard was exceeded most frequently at the Winnebago River, Iowa (07080203), where exceedance occurred approximately once every 6 days.

Over the 42-year modeling period, the average nitrate yield of all subbasins was 6.2 ± 4.6 kg-N/ha/yr. Generally, the largest mean yields were in the Midwest, such as at the West Fork Cedar River in Iowa (21.0 kg-N/ha/yr; 07080204), and the Vermillion River in Illinois (20.8 kg-N/ha/yr; 05120109). The lowest nitrate yields occurred in basins in North Dakota, which drain toward the Missouri River, such as the James River (0.02 kg-N/ha/yr; 10160001) and the West Missouri Coteau Watershed (0.03 kg-N/ha/yr; 10130106). The largest interannual variations of nitrate concentrations and yields occur primarily in the Midwest (Figure 7). Average nitrogen surplus in the MRB was 36.7 ± 18.3 kg-N/ha/yr (Byrnes et al., 2020), with the largest surpluses in the Midwest (>60 kg-N/ha/yr) and the lowest (<20 kg-N/ha/yr) in the Appalachian Mountains.

The occurrence of large-scale hydrologic extremes, such as regional floods and droughts, had considerable impact on the spatial gradients of modeled flow and nitrate (Figure 8). Maps of mean annual flow, nitrate concentration, and nitrate yield are shown for three archetypal years of wetness (represented by mean annual precipitation in the MRB): a dry year (980 mm; 2012), a typical year (1,091 mm; 2007), and a wet year (1,259 mm; 1993). Unsurprisingly, larger flow and nitrate export occurs during wetter years. In the Midwest region, flow magnitudes vary considerably from the dry year (2012) to the wet year (1993). On the other hand, flow in the Ohio River Vally is more stable across the 3 years. Nitrate concentrations are greatest in the Midwest, particularly in northern Iowa and southern Minnesota. Concentrations are dependent on annual precipitation and thereby watershed wetness conditions, with higher concentrations during wet years across most of the MRB (Figure 8). Nitrate yield was most influenced spatially by the distribution of nitrate concentrations and temporally by the interannual variability in flow. In the wet year (1993), the mean nitrate yield in the MRB was 9.7 kg-N/ha/yr with high yields (56.7 kg-N/ha/yr) in locations like Blue Earth River, Minnesota (07020009). In the typical (2007) and dry (2012) years, mean nitrate yields in the MRB were 6.1 kg-N/ha/yr and 3.3 kg-N/ha/yr, respectively. At Blue Earth River, typical and dry year yields were 10.8 kg-N/ha/yr and 4.3 kg-N/ha/yr, respectively, highlighting the influence of interannual hydrologic variability on nitrogen export.

The NDR is a simple metric to assess the relative export of surplus nitrogen from a watershed (Figure 7). Average NDR in the MRB was calculated to be 0.19 ± 0.04, supporting the prevailing paradigm that the vast majority of surplus nitrogen is not exported by riverine pathways, but rather it is retained on the landscape, to be denitrified or leached. These results agree with recent retention estimates ($R = 1-\mathrm{NDR}$) of approximately 80% over the same period (Stackpoole et al., 2021). Higher NDR values can be observed both in locations with large nitrogen surpluses (e.g., Iowa) as well as locations with lower surpluses (e.g., West Virginia), depending on the balance of surplus versus export.

The interannual (or temporal) variations in regional wetness influence NDR considerably (Figure 8). Mean NDR values across MRB for dry, typical, and wet years were 0.12, 0.19, and 0.24, respectively. Interannual variations were greatest in Midwestern states, such as Iowa where relative export of nitrate surplus was much higher in wet years (0.42 ± 0.24) compared to dry years (0.06 ± 0.06). This is primarily driven by large differences in streamflow in Iowa from the wet year (1.21 ± 0.42 mm/d) to the dry year (0.34 ± 0.19 mm/d). In the eastern regions of the MRB, where interannual variation in rainfall is lesser, so is the variation in NDR (Figure S8 in Supporting Information S1). As an example, for basins in West Virginia, the gap between NDR in the wet year

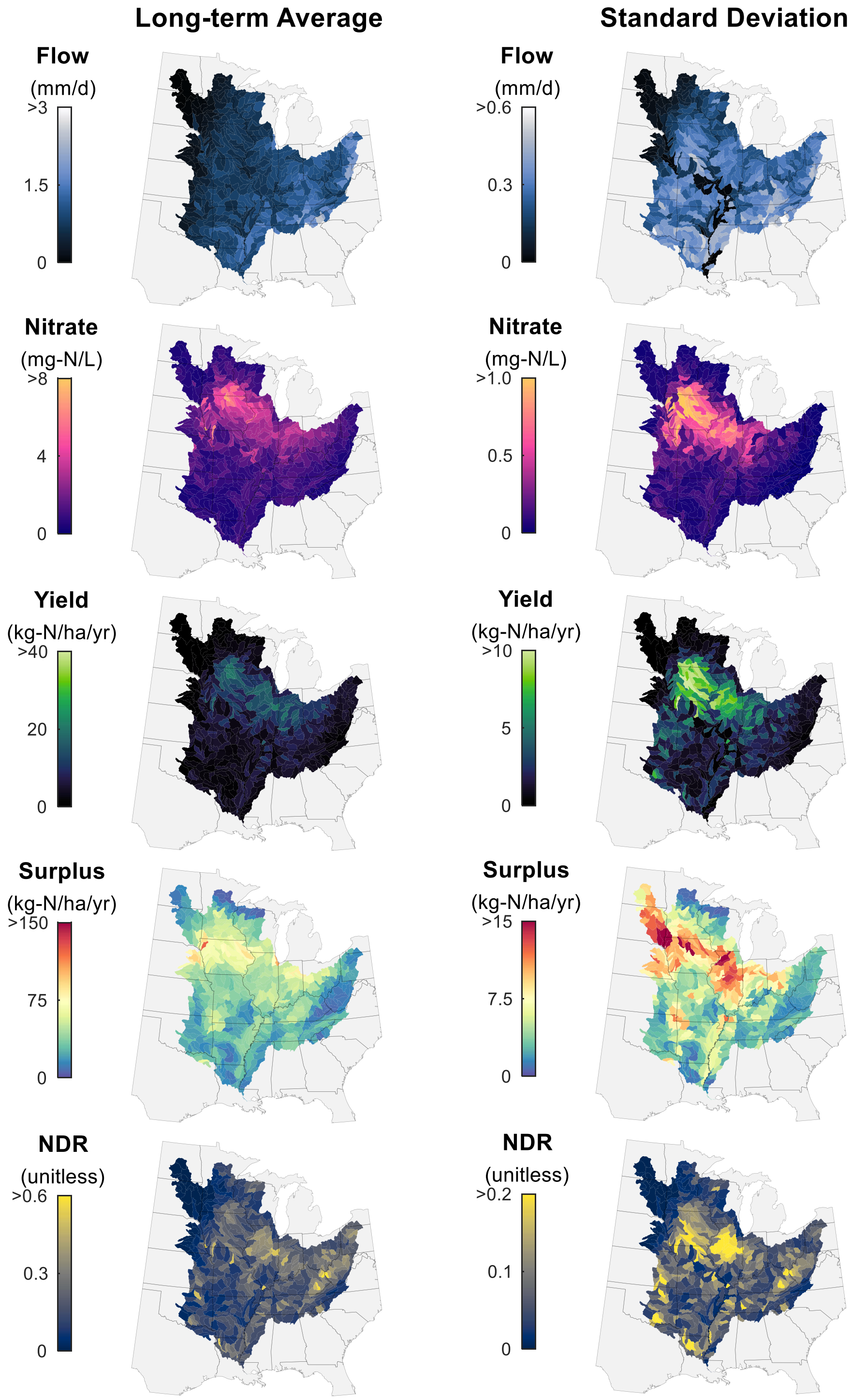


**Figure 7.** Model results for long-term mean and standard deviation of flow, nitrate concentration, and yield (from 1980 to 2022) and nitrogen surplus and Nitrogen Delivery Ratio (from 1980 to 2017) in the MRB study domain. NDR, Nitrogen Delivery Ratio.

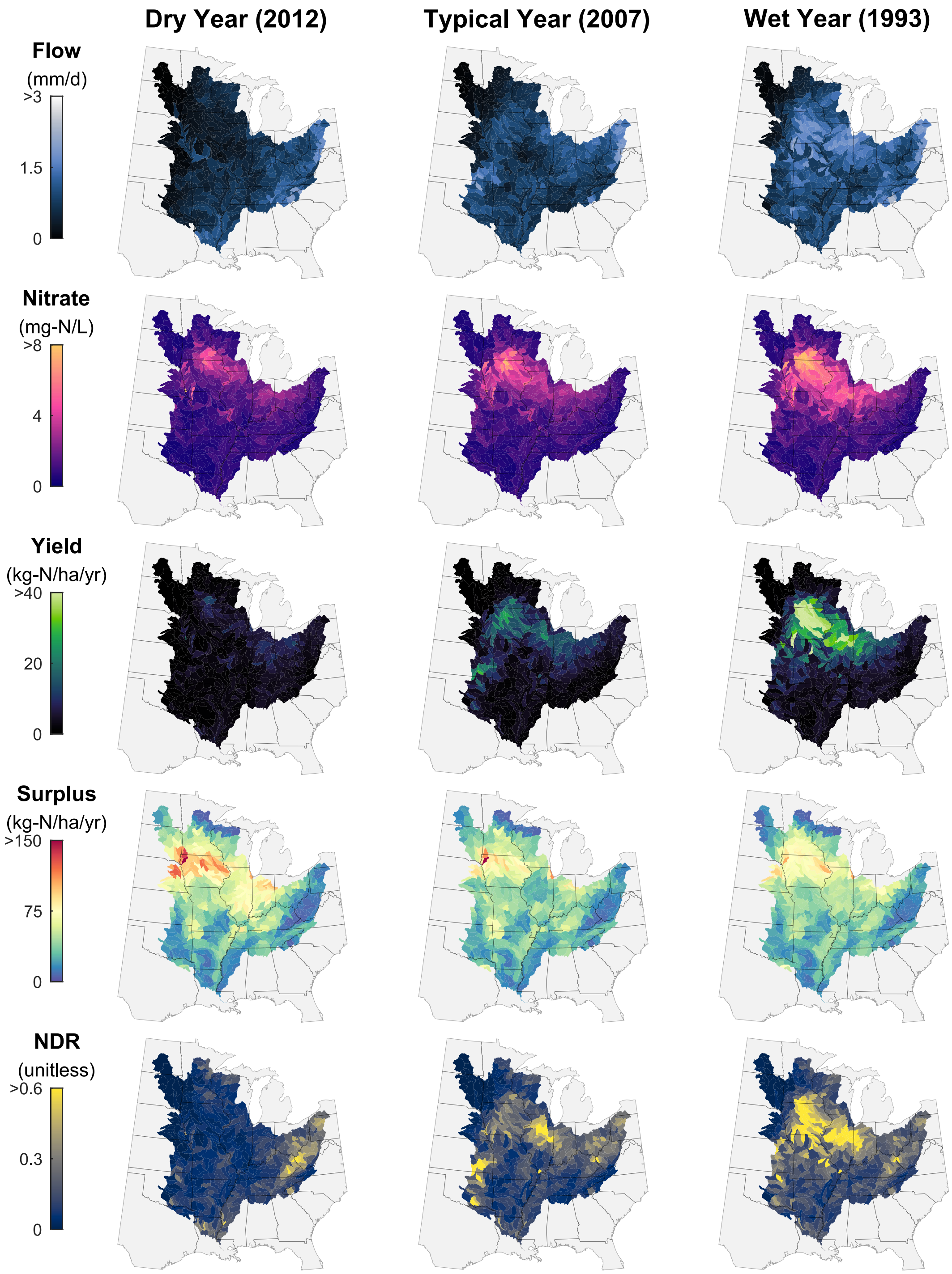


**Figure 8.** Model results for a dry year (2012), a typical year (2007), and a wet year (1993) in the MRB study domain. The upper Midwestern states, like Illinois, Iowa, and Minnesota, experience the greatest sensitivity of nitrate yield and delivery in response to regional-scale hydrology. NDR, Nitrogen Delivery Ratio.

(0.26 ± 0.14) and the dry year (0.37 ± 0.21) is smaller than that of Iowa. This is due in part to the relative stability in streamflow from the wet year (1.08 ± 0.34 mm/d) to the dry year (1.04 ± 0.28 mm/d) in West Virginia.

## 4. Discussion

### 4.1. Deep Learning for Nitrate Prediction

#### 4.1.1. Temporal Forecasting of Nitrate

We present a robust application of an LSTM model, trained using sensor data, for the large-scale prediction of daily nitrate concentrations and yields. For the temporal nitrate concentration prediction task, median KGE, NSE, and $R^2$ values for sites within the Mississippi River Basin (MRB) were 0.69, 0.57, and 0.59, respectively (Figure 4). These results indicate that the model has "good" temporal forecasting ability based on established metrics for daily nitrate evaluation (Pandit et al., 2025). Our model compares favorably to LSTM nitrate models developed at smaller scales where systems may exhibit less heterogeneity in climate, topography, soils, geology, and land use. Saha et al. (2023) simulated daily nitrate concentration for 42 sites in Iowa, USA, where their model achieved a higher performance than ours (median KGE and NSE of 0.77 and 0.63, respectively). However, if we only consider the 11 sites in our analysis located in Iowa, the median KGE and NSE values for our model are 0.81 and 0.73, respectively, highlighting the advantages of global model training. Gorski et al. (2024) developed an LSTM model—trained on sensor data from sites located primarily in the Midwest—to predict the 7-day mean nitrate concentration (median KGE of 0.63, $n = 46$). Our 1-day nitrate concentration model for a broader variety of sites achieved similar predictive performance (median KGE of 0.60, $n = 95$). Lastly, at the continental-scale, our model evaluated at all 95 sites (median KGE = 0.60) outperforms two LSTM (median KGE = 0.55) and WRTDS (median KGE = 0.47) models trained on grab samples from 131 sites across CONUS (Fang et al., 2024). In particular, Fang et al. (2024) note that "to fully assess the potential of the LSTM model, it may be necessary to use a higher frequency data set across the CONUS, which does not exist today." While the sensing data set we utilize does not have balanced coverage across the full US (see Figure 1), the improvements conferred to model accuracy by the data set underscores the point that high frequency data are crucial for training robust models.

#### 4.1.2. Prediction in Unmonitored Basins

Spatial validation—or prediction in unmonitored basins—has become a standard tool for assessing large-scale deep learning models of hydrology (Heudorfer et al., 2025). By comparison, the adoption of spatial validation of deep learning models of water quality has been underdeveloped with most studies conducting only temporal validation (Willard et al., 2025). While model performance in temporal forecasting at a site used in training is undoubtedly important, the true test of a deep learning model is its ability to generalize in space (Arsenault et al., 2023). Two recent daily sediment concentration models were some of the first to apply spatial validation to continental-scale LSTM model predictions (Gupta & Feng, 2025; Song et al., 2024). In both studies, the authors showed a considerable drop ($\Delta$NSE $\approx -0.20$) in the performance of the median site between temporal and spatial validation, owing to a lack of data to generalize information. We similarly saw drops in model performance for spatial validation task. For flow, this drop was modest for the MRB ($\Delta$KGE $= -0.11$; Figure 4) and CONUS ($\Delta$KGE $= -0.13$; Figure S4 in Supporting Information S1) as there are hundreds of stream gages with relatively balanced spatial coverage (Figure 1), allowing the model to learn general representations. For nitrate concentration, this drop was larger for the MRB ($\Delta$KGE $\approx -0.35$; Figure 4) and CONUS ($\Delta$KGE $\approx -0.42$; Figure S4 in Supporting Information S1), due to fewer sites for training, shorter data records, and an imbalance in spatial coverage. While these challenges limit the ability of current models, they also provide an opportunity for strategically collecting data (Zhi et al., 2024). Thus, we recommend that modelers assess their deep learning models of water quality with spatial validation techniques, in addition to temporal validation, to improve transparency regarding model generalizability.

#### 4.1.3. Aquatic Sensors as Big Data

High resolution aquatic sensing has numerous advantages over traditional discrete grab sampling for understanding nitrate dynamics (Jones, Davis, et al., 2018; Rode et al., 2016). We argue that an additional advantage is the large volume of data contained within aquatic sensing time-series, which will be crucial for training "data hungry" machine learning models (Lu et al., 2023). As an example, a recent benchmark data set for catchment geochemistry, termed CAMELS-Chem (Sterle et al., 2024), contains 22,320 grab sample observations of nitrate from 429 catchments over a 70-year period (1930–2020). This results in approximately 0.6 samples/year/catchment. On the other hand, the nitrate sensing data set used in this study ($n = 95$) contains 158,282 daily-averaged observations over a 15-year period (2007–2022). This corresponds to 111 samples/year/catchment, nearly two orders of magnitude of additional observations. A limitation of aquatic sensors is that they require a large upfront cost for installation. Thus, they are often installed in locations where nitrate is a major pollutant, biasing coverage in places such as the Midwest, leaving large geographical areas unmonitored. Nonetheless, considerable investments in aquatic sensing have been made by various academic, government, non-profit, and industry groups; however, these data are stored and maintained disparately from one another (Bieroza et al., 2023). To this end, there is a need to develop sensor data repositories to unite the swaths existing sensor data to fully realize the potential of continental-scale deep learning. Furthermore, while high-resolution data and model products are valuable, translating them into effective policy is complex due to limited adoption and difficulty in targeting interventions effectively (Del Rossi et al., 2023). The challenge, therefore, lies not only in acquiring high-resolution data but also in effectively integrating it into policy frameworks that address the complex economic and social factors influencing agricultural practices.

#### 4.1.4. Model Limitations and Future Work

Despite the strong performance of the LSTM models, there are opportunities for further refinement. First, additional data sources beyond sensors, such as integrating grab samples (as in Saha et al. (2023)), could be included in the model training process to leverage all available data. In particular, data coverage is needed in sparsely monitored locations, like much of the western US, to develop models that can extrapolate to those regions. The inclusion of high resolution dynamic forcings, such as dynamic land use change, best management practices (BMPs), or daily wastewater effluent data, could improve performance although these variables are difficult to aggregate at the continental scale. In general, latent variables exist that may exert considerable control over nitrate dynamics but are not explicitly characterized in the input data set. Additionally, at present, all catchment attributes are assumed to be constant (static) over the period of study, which may not represent the land use change occurring in some basins. Regarding the attributes, there is also uncertainty associated with how the properties in the reference data sets were calculated, such as the nitrogen surplus estimates derived from TREND (Byrnes et al., 2020). While our focus was on daily prediction, future work should look to leverage the potential of aquatic sensors for training models to make sub-daily predictions. Sensors typically collect sub-hourly data, thus there is an unrealized potential for sensor data to simulate high temporal frequency phenomena, like concentration-discharge patterns (Knapp et al., 2020). Beyond increased computational cost for sub-hourly prediction, another challenge to this end is that the lower bound for time resolution in a model is specified by the meteorological model forcings. While not broadly available at the present, future high temporal resolution climate forcing data may facilitate machine learning models to learn fine-scale water quality patterns (Ortiz-Lopez et al., 2022).

### 4.2. Revealing Spatial Drivers for Management

This study leverages the power of explainable machine learning to not only predict nitrate concentrations but also to unravel the complex interplay of factors driving these predictions. Our analysis captured a hierarchy of physical and anthropogenic influences that is broadly consistent with decades of process understanding (Figure 5). Climate and soil moisture variables, like precipitation, aridity, and soil water content, dominate spatial variability in flow, whereas predictors associated with nitrogen loading and hydrologic connectivity—tile-drainage extent, roadway density, wetland cover, sand fraction, and precipitation—rise to the top for nitrate concentration. This agreement between process knowledge and data-driven importance gives us confidence that the RF model is not merely relying on spurious correlations. Beyond confirming well-understood drivers, our study adds to this body of knowledge by revealing non-linear thresholds where small changes in a feature can lead to substantial shifts in predicted nitrate levels (Figure 6). By understanding where (spatially) and at what level (thresholds) certain landscape features become critical influencers of nitrate, interventions can be designed with greater precision and potential for success.

From a management perspective, these inflection points translate into actionable guidelines for screening basins for remediation. Basin screening could prioritize (1) tile-drained watersheds above roughly 20% drainage density; (2) karst landscapes with greater than 40% carbonate coverage; and (3) urbanizing basins exceeding a roadway density of 0.8 $km/km^2$ (Figure 6). Each of these feature thresholds are related to increased longitudinal hydrologic connectivity and reduced residence time, which together short-circuit natural bioremediation and uptake processes that would reduce nitrate concentrations (Covino, 2017). Conversely, our results lend quantitative support to wetland restoration targets: increasing wetland area to above 1% could shift $\psi_{\text{wetland cover}}$ far enough negative to partly offset the nitrate increase imposed by tile drains ($\psi_{\text{tile drainage}}$) or other drivers. In this way, wetlands reduce longitudinal connectivity, limiting the pulses of nitrate, and increase lateral connectivity to redox conditions in the landscape that aid in nitrate reduction (Jensen & Ford, 2019). While our modeling approach provides thresholds that are spatially integrated over entire basins, it is important to note that the within-basin spatial configuration of wetlands can be a controlling factor in their efficacy for nitrate removal (Evenson et al., 2021). Beyond wetland cover, the continental-scale data sets used in this study (see Table S1 in Supporting Information S1) do not have estimates of other types of BMPs, thus we are unable to reveal how other conservation practices influence nitrate concentrations. Future work could aggregate data sets that encode for these influence and apply the approach described herein to identify the relative effect of various conservation practices. Lastly, while certain important features are not manageable, such as precipitation and sand fraction (Figure 5), the quantification of these drivers to influence nitrate concentrations is nonetheless important physical context that can influence management decisions in various landscape settings.

Leveraging the insights from our interpretability approach, our findings suggest returning to past calls for optimizing agricultural practices, including in-field practices such as fertilizer application and cover crops (Williams et al., 2015), as well as edge-of-field practices such as intercepting tile drainage with control structures and wetlands (Crumpton et al., 2020). Our results support the notion that wetlands, even when they cover a relatively small footprint (~1%) can confer substantial benefits to nitrate reduction (Hansen et al., 2021). The combination of these suites of practices are needed to limit nitrate delivery to rivers (David et al., 2010). Karst landscapes have elsewhere been identified to cause threshold increases in nitrate concentration, at approximately the same coverage (~40%; Wherry et al., 2021), and should be managed to reduce nitrogen inputs as karst features can bypass natural attenuation processes (Husic et al., 2019, 2020). In urban systems, we identify that even modest roadway densities can exert control on nitrate levels (e.g., 0.8 km/km$^2$, Figure 6), echoing similar findings (Decatanzaro et al., 2009). Frequent, small storm events can generate runoff on these impervious roadways, thus urban management can benefit from prioritizing runoff capture with vegetated strips or flow-control structures (Boger et al., 2018; Shrestha et al., 2018). Sustainable development practices will be essential in mitigating the adverse effects of urbanization and agricultural expansion. We recognize this is a long-standing and formidable challenge. However, continued efforts to reduce nitrate export in the water system are important, and the results presented by this study can inform both the temporal and spatial components of that effort.

Lastly, it is important to consider that the explanations made by the Shapley value analysis are a function of (1) the set of basins and catchment attributes considered, in this case primarily agricultural basins with drivers that were accessible via continental-scale data sets; (2) the choice of model, in this case a random forest; and (3) how accurate the model is, in this case fairly accurate ($R^2 \geq 0.57$, Table S2 in Supporting Information S1) (Husic, 2025a). With particular focus on point 1, if a model were constructed to explain nitrate for 100 pristine forested sites, it may have vastly different drivers than a model for explaining 100 tile-drained agricultural sites. The exact thresholds and drivers will vary as a function of the sites included in the analysis, as demonstrated by a recent analysis of several hundred minimally impacted basins (Husic, 2025a). Thus, our results are applicable to the sites considered in our analysis, which are primarily concentrated in the Midwest region and the Chesapeake Bay Basin. The average basin in this data set is characterized by several attributes that contribute to high riverine nitrate, including extensive crop cover (44%), tile drainage (16%), and urban cover (12%). Regarding other physical landscape features, karst extent was 27%, forest cover was 37%, and wetland cover was 7%. Lastly, there may be latent factors that are not directly included in our analysis (e.g., wastewater effluent) but are represented by other factors (e.g., urban cover, population density, and roadway density). The explicit inclusion of the latent factor itself would aid in distinguishing its contribution versus that of a covariate. Thus, while Shapley values can enhance model transparency, the resulting interpretations must be tempered by the scope and accuracy of the model and the completeness of the input data.

### 4.3. Nitrate Export in the Mississippi River Basin

This study provides a valuable new data set of modeled daily flow, nitrate concentration, nitrate yield, and annual nitrate surplus export for the majority of HUC-8 basins in the MRB (see Data Availability Statement), which we

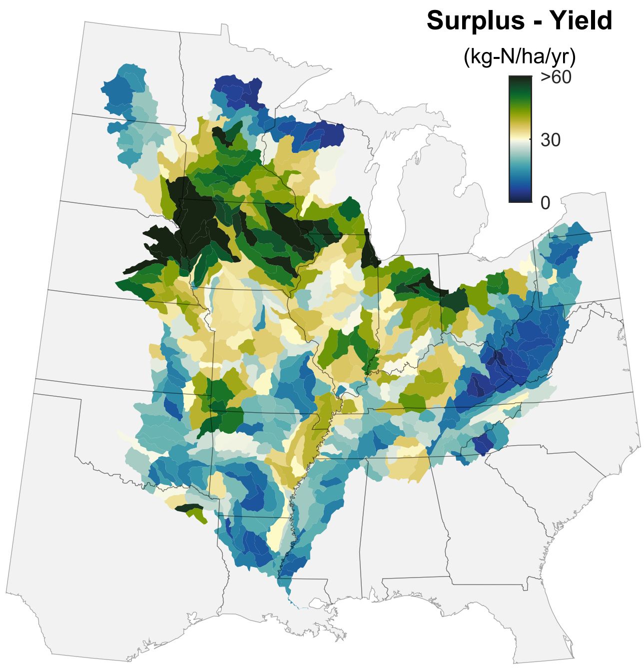


**Figure 9.** Model results for long-term (1980–2017) mean nitrogen surplus minus nitrate yield in the MRB study domain.

use to assess patterns of disproportionate nitrate loading. Over the 42-year study period, the 100 highest yield basins, representing just 18% of the drainage area, exported 43% of the total nitrate load. During the particularly wet year of 1993, these same basins accounted for 53% of nitrate load, highlighting the amplifying role of precipitation in interannual nitrogen loading (Baeumler & Gupta, 2020; Sinha & Michalak, 2016). The finding that a small number of basins dominate nitrate export is consistent with previous studies, such as in the broader Mississippi-Atchafalaya Basin, where selected basins in Iowa have been estimated to contribute between 11% and 52% of the nitrate load despite covering only 4.5% of the drainage area (Jones, Nielsen, et al., 2018). In our analysis, basins draining at least part of Iowa ($n = 56$) export between 13% and 33%, despite covering 12% of the study domain. Because our study domain includes only the HUC-8 basins east of the 98th meridian (Figure 2), covering approximately 60% of the Mississippi-Atchafalaya Basin, our modeled estimates likely understate the true degree of disproportionate loading. Although they account for 40% of the drainage area, basins west of the 98th meridian contribute only about 15% of the total nitrate load (Crawford et al., 2019), meaning that disproportionate loading by eastern basins is likely even more extreme than captured here.

Not all surplus nitrogen in a basin is exported by riverine pathways. We show that the long-term relative export of nitrate relative to surplus nitrogen (NDR) for basins in the MRB is around 0.19 ± 0.04 (Figure 7). David et al. (2010) similarly found that the long-term ratio of net nitrogen inputs to riverine export ranged from 0.05 in the lower MRB to 0.30 in the upper MRB, generally matching the spatial patterns we observe. As NDR is a ratio of yield-to-surplus, high delivery ratios typically occur in places with (relatively) large yields and low surpluses (Figure S9 in Supporting Information S1). In West Virginia, where nitrogen surpluses are low (<20 kg-N/ha/yr) and hydrology contributes to consistent export, long-term mean NDR values are high (0.30 ± 0.16, $n = 25$). In a study of sixteen forest-dominated catchments in the Northeastern US, where atmospheric deposition was the main source of nitrogen, Boyer et al. (2002) similarly found that riverine exports accounted for 25% of inputs. However, large values of NDR can also occur in high surplus areas (>60 kg-N/ha/yr), if the additional surplus nitrogen is readily exportable, as demonstrated in Iowa basins (0.20 ± 0.11, $n = 56$). Despite the similar ratios of nitrate export between agricultural and forested basins, the mass of excess nitrogen remaining on the landscape (nitrogen surplus minus yield; Figure 9) is considerably greater in agricultural basins in Iowa (49.7 ± 17.0 kg-N/ha/yr) compared to forested basins in West Virginia (10.0 ± 5.1 kg-N/ha/yr).

Interannual variability in regional wetness considerably impacts the relative export of surplus nitrogen. This is most apparent in the Midwest, such as for Iowa basins where the NDR was much higher in the wet year (0.42 ± 0.24) compared to the dry year (0.06 ± 0.06) (Figure 8). The higher relative export in wet years could be due, in part, to greater hydrologic connectivity, allowing for remobilization of a given year's nitrogen surplus (Husic et al., 2023). Further, several basins exhibit NDR values greater than one—that is, more nitrate export than surplus in a year—indicating the potential export of legacy nitrogen that accumulated within the landscape from prior years (Van Meter et al., 2017, 2018). The NDR exceeded 1.0 on 48 occasions at 23 unique basins. At two basins, NDR exceeded 1.0 on six occasions: Twelvepole Watershed, West Virginia (05090102), and Red Chute Watershed, Louisiana (11140204). At these locations, low nitrogen surpluses combined with high flows result in high nitrogen delivery. At the other unique basins where NDR exceeded unity, 10 can be considered intensively agricultural (yield >10 kg-N/ha/yr). Of the 10, seven drained parts of Illinois, such as the Vermillion River (05120109, NDR = 1.10 in 2009) and Green River (07090007, NDR = 1.11 in 2009 and 1.02 in 2010).

Our modeling shows that nitrate export can exceed surplus in agricultural landscapes, highlighting the remobilization of legacy nitrogen stores. Van Meter et al. (2018) suggest that even if all of the fertilizer applied to agricultural lands was taken up by crops, it would still take decades to meet management targets due to large stocks of stored nitrogen, particularly in groundwater (Sprague et al., 2011; Van Meter et al., 2023). Further, in the Midwest, the abrupt transition from a dry year to a wet year has been shown to elevate nitrate concentrations in rivers in a process termed "weather whiplash" (Loecke et al., 2017), whereby accumulated nitrogen from 1 year is flushed in the next (Loecke et al., 2017). Our analysis supports this idea as interannual changes to MRB nitrate concentrations in wet years following dry years ($n = 7$) were significantly larger than interannual variations between all other years ($n = 23$; $p < 0.05$). Our model shows that every 1 mm/d increase in flow, from 1 year to the next in the MRB, is accompanied by a 0.85 mg-N/L increase in nitrate concentration ($R^2 = 0.71$; Figure S10 in Supporting Information S1). Together, these lines of evidence lend credence to the hypothesis that nitrate legacy stores from prior years constitute an apparent greater magnitude of surplus nitrate export during wet years.

Effective nitrogen management requires understanding both the spatial patterns and temporal dynamics of nutrient loss. Monitoring the relationship between surplus and yield whether via the NDR (Figure 8) or the absolute difference (Figure 9) can help not only identify catchments where excess nitrogen enters rivers but where more nitrogen is either remaining on the landscape or lost via other pathways (e.g., denitrification). By focusing efforts on systems that disproportionately contribute to the total nitrate yield, such as those in the upper Midwest, the greatest benefits can be realized. Crop rotations, shifts in the timing of fertilization, improved drainage system design, and off-field practices that promote denitrification, such as tile bioreactors and wetlands, may be implemented to balance production and water quality needs (Castellano et al., 2019; Crumpton et al., 2020). While monitoring the surplus-yield relationship identifies where management should be prioritized, our LSTM approach provides crucial insights into when proactive management strategies should be timed. For example, using the current state of the system—such as inferred soil moisture conditions and estimated stored nitrogen, as captured by the LSTM—and anticipated meteorologic conditions, forecast simulations can inform decisions, such as delaying fertilizer application, if high losses are projected, or triggering adaptive BMPs, such as drainage-water management structures (Rittenburg et al., 2015; Sangha et al., 2023; Tamagno et al., 2022). This capacity for dynamic forecasting opens possibilities for more adaptive and efficient nutrient management frameworks.

## 5. Conclusions

Accurately predicting riverine nitrate export, particularly its high temporal and spatial variability within basins like the MRB, remains a critical challenge for managing downstream eutrophication. This study leveraged advances in high-frequency aquatic sensing and machine learning to address this challenge. We developed an LSTM model that achieved strong performance in temporal predictions of daily nitrate concentration (e.g., median KGE 0.69 in MRB), demonstrating the power of high-frequency data for capturing nitrate dynamics. However, rigorous spatial cross-validation revealed significant challenges in generalizing these predictions to unmonitored basins in the MRB (median KGE dropped to 0.34), underscoring the limitations imposed by current data network imbalances and the need for careful spatial assessment of model transferability. Complementing the temporal model, an interpretable Shapley value approach elucidated the key drivers of spatial variation in long-term nitrate metrics, confirming the influence of factors like tile drainage, wetland cover, and anthropogenic indicators, and crucially, identifying potentially actionable non-linear thresholds in their influence.

Applying the LSTM model to the MRB (1980–2022) via domain adaptation provided insights into basin-scale nitrate yield variation. Our results quantify the low overall relative export of excess nitrate from the MRB (NDR = 0.19), confirming the large-scale retention of surplus nitrogen and highlighting the potential importance of legacy stores. We demonstrated strong interannual variability in export, driven by climate fluctuations, with NDR values significantly higher in wet years compared to dry years, particularly in the agricultural Midwest (e.g., Iowa NDR ∼42% vs. ∼6%). Evidence, including NDR values exceeding unity, supports the hypothesis that legacy nitrogen remobilization contributes significantly to export during wet periods. Furthermore, our analysis confirmed disproportionate nitrate loading from specific hotspot subbasins, emphasizing the spatial heterogeneity of nutrient sources. Together, these findings provide a temporally dynamic and spatially explicit understanding of nitrate export, offering valuable information for refining nutrient management strategies, such as targeting interventions based on identified landscape thresholds, focusing efforts on hotspot basins, and accounting for climate variability and legacy nitrogen impacts in efforts to mitigate Gulf of Mexico hypoxia.

## Data Availability Statement

The codebase for the LSTM model is available through the HydroDL repository (Feng et al., 2020). The sensor training data, model outputs, and performance evaluation metrics for the LSTM models are hosted on the Open

Science Framework (Husic, 2025b). The daily model predictions of flow, nitrate concentration, and nitrate yield for the 505 HUC-8 basins in the MRB are also hosted on the Open Science Framework (Husic, 2025b).

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